

St3g0

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Download this image and find the flag.

Hints

- We know the end sequence of the message will be `$t3g0`.
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Tools Used

- `file` – Identify image file type and properties
 - `strings` – Extract readable text from binary files
 - `exiftool` – Inspect image metadata and embedded text
 - `binwalk` – Detect embedded files and hidden data
 - `zsteg` – Detect steganography in PNG/BMP images (LSB analysis)
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Complete Solution

Step 1: Download and Prepare

Download the image named `pico.flag.png` from the challenge page.



At first glance, this appears to be a simple PNG image with a lock icon.

Step 2: Basic File Inspection

First, verify the file type and general information:

```
file pico.flag.png
```

Expected output:

```
pico.flag.png: PNG image data, 585 x 172, 8-bit/color RGBA, non-interlaced
```

The image is a valid PNG with dimensions 585×172 pixels and an alpha channel (RGBA).

Step 3: Search for Visible Text

The flag may be present as plain text. Let's try extracting all character strings:

```
strings pico.flag.png | grep -i "picoCTF"
```

Output: (no results)

⚠ No flag appears yet. We need to dig deeper.

Step 4: Inspect Metadata

Use `exiftool` to examine the file metadata for any hidden clues:

```
exiftool pico.flag.png
```

Complete output:

```
ExifTool Version Number      : 13.50
File Name                    : pico.flag.png
Directory                   : .
File Size                    : 13 kB
File Modification Date/Time  : 2026:05:10 18:20:17+02:00
File Access Date/Time       : 2026:05:10 18:20:55+02:00
File Inode Change Date/Time  : 2026:05:10 18:20:48+02:00
File Permissions             : -rw-rw-rw-
File Type                    : PNG
File Type Extension          : png
MIME Type                    : image/png
Image Width                  : 585
Image Height                 : 172
Bit Depth                   : 8
Color Type                   : RGB with Alpha
Compression                  : Deflate/Inflate
Filter                       : Adaptive
Interlace                    : Noninterlaced
Image Size                   : 585x172
Megapixels                   : 0.101
```

The metadata itself is not suspicious.

Step 5: Scan with Binwalk

Use `binwalk` to check for any embedded files:

```
binwalk pico.flag.png
```

Output:

DECIMAL	HEXADECIMAL	DESCRIPTION

0	0x0	PNG image, 585 x 172, 8-bit/color RGBA, non-interlaced
41	0x29	Zlib compressed data, default compression

Nothing suspicious here either - no embedded ZIP or other files.

Step 6: Analyze Steganography with zsteg

Since the image is an **8-bit RGBA PNG**, it may contain hidden data in the **least significant bits (LSB)** of its pixels. This is a common steganography technique.

We use `zsteg`, a tool specifically designed to detect LSB steganography in PNG and BMP images:

```
zsteg pico.flag.png
```

Output:

```
b1,r,lsb,xy      .. text: "~__B>wV_G@"
b1,rgb,lsb,xy    .. text: "picoCTF{<REDACTED>}$t3g0"
b1,abgr,lsb,xy   .. text: "E2A5q4E%uSA"
b2,b,lsb,xy      .. text: "AAPAAQTAAA"
b2,b,msb,xy      .. text: "HWUUUUUUU"
b4,r,lsb,xy      .. file: Targa image data (16-273) 65536 x 4097 x 1 +4352 +4369 -
1-bit alpha - right "\021\020\001\001\021\021\001\001\021\021\001"
b4,g,lsb,xy      .. file: 0420 Alliant virtual executable not stripped
b4,b,lsb,xy      .. file: Targa image data - Map 272 x 17 x 16 +257 +272 - 1-bit
alpha "\020\001\021\001\021\020\020\001\020\001\020\001"
b4,bgr,lsb,xy    .. file: Targa image data - Map 273 x 272 x 16 +1 +4113 - 1-bit
alpha "\020\001\001\001"
b4,rgba,msb,xy   .. file: Applesoft BASIC program data, first line number 8
```

Key finding: The line `b1,rgb,lsb,xy` reveals the flag!

The extracted text is:

```
picoCTF{<REDACTED>}$t3g0
```

The hint mentioned that the message ends with `$t3g0`, which confirms we have found the correct hidden data.

Step 7: Verify the Flag

The flag is embedded directly in the LSB of the RGB channels. No further decoding is needed.

Final Result

picoCTF{<REDACTED>}

Note: The actual flag is redacted here. In the real challenge, `zsteg` will display the complete flag.

Key Concepts

- **Steganography:** The practice of hiding data within other non-secret data (e.g., hiding a message in the least significant bits of image pixels).
 - **LSB (Least Significant Bit):** The last bit of a byte. Changing it has minimal visual impact, making it ideal for hiding data.
 - **RGBA Image:** A PNG image with Red, Green, Blue, and Alpha (transparency) channels. Each channel can hide data.
 - **zsteg:** A tool that detects LSB steganography in PNG and BMP images by analyzing pixel channels and bit planes.
 - **Metadata Analysis:** Using `exiftool` and `binwalk` to rule out other hiding methods (metadata, appended files).
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Additional Resources

You can also use online tools to analyze steganography in images:

- [AperiSolve](#) – Online steganography analysis tool